

Hans van Gorp

Postdoctoral Researcher · TU/e, NXP Semiconductors

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Research areas

- Signal Processing
- Deep learning
- Deep generative models
- Deep learning for inverse problems
- Active Inference
- Automatic sleep staging
- FMCW radar processing
- Ultrasound operator guidance

Education

- PhD (cum laude) on deep generative modeling in sleep diagnostics, Eindhoven University of Technology and Philips Sleep and Respiratory Care, 2026
- MSc in Electrical Engineering, Eindhoven University of Technology, 2020
- BSc in Electrical Engineering, Eindhoven University of Technology, 2018

Work experience

- February 2026 - Present: Postdoctoral researcher
 - Eindhoven University of Technology, Department of Electrical Engineering, Signal Processing Systems
 - NXP Semiconductors
 - Project: Deep learning in automotive radar
- September 2020 - January 2026: PhD (cum laude)
 - Eindhoven University of Technology, Department of Electrical Engineering, Signal Processing Systems, BM/d Lab
 - Philips Sleep and Respiratory Care, Eindhoven
 - Project: Deep Generative Modeling in Sleep Diagnostics
- July 2024 - November 2024: PhD Internship
 - Qualcomm AI Research, Amsterdam
 - Project: Neural Augmented Kalman Filters for Road Network Assisted GNSS Positioning
- August 2019 - October 2019: Research Intern
 - Philips Research, Eindhoven
 - Project: Suppression of pump distortion for inflation-based noninvasive blood pressure measurement

Grants awarded

- 2026: TTT-AI voucher
 - Polaris: Operator Guidance for Carotid Ultrasound

Publications

- **Ultrasound Operator Guidance Using World Modeling and Retrieval Based Action Planning**
Noortje I.P. Schueler, **Hans van Gorp**, and Ruud J.G. van Sloun. *arXiv preprint*, 2026.
- **The Geometry of LISTA**
Hans van Gorp, Oisín Nolan, Yonina C Eldar, and Ruud J G van Sloun. *IEEE Signal Processing Magazine*, 2026.
- **Automatic Sleep Staging and Atonia Detection in RBD Using Single-Channel Chin EMG**
Hans van Gorp, Jaap F van der Aar, Ruud J G van Sloun, Angelique Pijpers, Pedro Fonseca, Sebastiaan Overeem, Merel M van Gilst. *Journal of Parkinson's Disease*, 2026.
- **Deep Generative Modeling in Sleep Diagnostics**
Hans van Gorp. *PhD thesis, Eindhoven University of Technology*, 2026.
- **A Deep Generative Model for Five-Class Sleep Staging with Arbitrary Sensor Input**
Hans van Gorp, Merel M van Gilst, Pedro Fonseca, Fokke B van Meulen, Johannes P van Dijk, Sebastiaan Overeem, and Ruud J G van Sloun. *IEEE Journal of Biomedical and Health Informatics*, 2025.
- **Neural Augmented Kalman Filters for Road Network Assisted GNSS Positioning**
Hans van Gorp, Davide Belli, Amir Jalalirad, Bence Major. *ICML Workshop on Machine Learning for Wireless Communication and Networks*, 2025.
- **SSC-SleepNet: A Siamese-Based Automatic Sleep Staging Model with Improved N1 Sleep Detection**
Songlu Lin, Zhihong Wang, **Hans van Gorp**, Mengzhu Xu, Merel M van Gilst, Sebastiaan Overeem, Jean-Paul Linnartz, Pedro Fonseca, Xi Long. *IEEE Journal of Biomedical and Health Informatics*, 2025.
- **Removing Structured Noise Using Diffusion Models**
Tristan S.W. Stevens, **Hans van Gorp**, Faik C. Meral, Junseob Shin, Jason Yu, Jean-Luc Robert and Ruud J.G. van Sloun. *Transactions on Machine Learning Research (TMLR)*, 2025.
- **Deep Sylvester Posterior Inference for Adaptive Compressed Sensing in Ultrasound Imaging**
Simon W Penninga, **Hans van Gorp**, and Ruud J G van Sloun. *ICASSP*, 2025.
- **Learning Structured Compressed Sensing with Automatic Resource Allocation**
Han Wang, Eduardo Pérez, Iris A. M. Huijben, **Hans van Gorp**, Ruud van Sloun, and Florian Römer. *ICASSP*, 2025.
- **Hypnogram and Hypnodensity Analysis of REM Sleep Behaviour Disorder Using Both EEG and HRV-Based Sleep Staging Models**
Jaap F. van der Aar, Merel M. van Gilst, Daan A. van den Ende, **Hans van Gorp**, Peter Anderer, Angelique Pijpers, Pedro Fonseca, Elisabetta Peri, Sebastiaan Overeem. *Journal of Sleep Research*, 2025.
- **The Tree of Diffusion Life: Evolutionary Embeddings to Understand the Generation Process of Diffusion Models**
Vidya Prasad, **Hans van Gorp**, Christina Humer, Anna Vilanova, and Nicola Pezzotti. *arXiv preprint*, 2024.
- **Single-Channel EOG Sleep Staging on a Heterogeneous Cohort of Subjects with Sleep Disorders**
Hans van Gorp, Merel M van Gilst, Sebastiaan Overeem, Sylvie Dujardin, Angelique Pijpers, Bregje van Wetten, Pedro Fonseca and Ruud J G van Sloun. *Physiological Measurement*, 2024.
- **Model-Based Diffusion for Mitigating Automotive Radar Interference**
Jeroen Overdevest, Xinyi Wei, **Hans van Gorp**, and Ruud J G van Sloun. *ICASSP*, 2024.
- **Studying Sleep: Towards the Identification of Hypnogram Features That Drive Expert Interpretation**
Caspar van der Woerd, **Hans van Gorp**, Sylvie Dujardin, Manuel Sastry, Humberto Garcia Caballero, Fokke van Meulen, Stef van den Elzen, Sebastiaan Overeem, Pedro Fonseca. *Sleep*, 2023.

- **Modeling the Impact of Inter-Rater Disagreement on Sleep Statistics Using Deep Generative Learning**
H. van Gorp, M. M. van Gilst, P. Fonseca, S. Overeem, and R. J. G. van Sloun. *IEEE Journal of Biomedical and Health Informatics*, 2023.
- **Aleatoric Uncertainty Estimation of Overnight Sleep Statistics Through Posterior Sampling Using Conditional Normalizing Flows**
H. van Gorp, M. M. van Gilst, P. Fonseca, S. Overeem, and R. J. G. van Sloun. *ICASSP*, 2023.
- **Certainty About Uncertainty in Sleep Staging: A Theoretical Framework**
H. van Gorp, I. A. M. Huijben, P. Fonseca, R. J. G. van Sloun, S. Overeem, and M. M. van Gilst. *Sleep*, 2022.
- **Deep Unfolding with Normalizing Flow Priors for Inverse Problems**
X. Wei, H. van Gorp, L. Gonzalez Carabarin, D. Freedman, Y. C. Eldar, and R. J. G. van Sloun. *IEEE Transactions on Signal Processing*, 2022.
- **Representations of Temporal Sleep Dynamics: Review and Synthesis of the Literature**
L. W. A. Hermans, I. A. M. Huijben, H. van Gorp, T. R. M. Leufkens, P. Fonseca, S. Overeem, and M. M. van Gilst. *Sleep Medicine Reviews*, 2022.
- **Image Denoising with Deep Unfolding and Normalizing Flows**
X. Wei, H. van Gorp, L. Gonzalez Carabarin, D. Freedman, Y. C. Eldar, and R. J. G. van Sloun. *ICASSP*, 2022.
- **Active Deep Probabilistic Subsampling**
H. Van Gorp, I. A. M. Huijben, B. S. Veeling, N. Pezzotti, and R. J. G. Van Sloun. *ICML*, 2021.
- **Dose Distribution as Outcome Predictor for Gamma Knife Radiosurgery on Vestibular Schwannoma**
P. P. J. H. Langenhuizen, H. van Gorp, S. Zinger, H. B. Verheul, S. Leenstra, and P. H. N. de With. *Medical Imaging*, 2019.

Patents

- **System and Method for Generating Output Data Using a Generative Model**
H. van Gorp, P. Fonseca, M. M. van Gilst, S. Overeem, and R. J. G. van Sloun. *US Patent Application 2025/0079004*, 2025.
- **System and Method for Estimating Sleep Stage**
H. van Gorp, P. Fonseca, M. M. van Gilst, S. Overeem, and R. J. G. van Sloun. *US Patent Application 2025/0072825*, 2025.

Teaching

- September 2026 - Present: Responsible lecturer
 - Eindhoven University of Technology
 - Course: Machine Learning for Signal Processing
- September 2020 - September 2026: co-lecturer
 - Eindhoven University of Technology
 - Course: Machine Learning for Signal Processing
- September 2016 - September 2020: Student Teaching Assistant
 - Eindhoven University of Technology
 - Courses: Computation, Calculus, Applied Physics

Volunteering

- Leader at Scouting Nederland for the 'welpen' (cub scouts), kids aged 7 through 11